

Atmospheric electricity

X24 (2024) — English translation

The phenomena of atmospheric electricity can be seen by anyone in everyday life. Each of us, although afraid, still admires the beauty of lightning appearing in the sky, or watches with delight the sparks in the air. As part of this task, you are invited to take a closer look at the electrical phenomena that occur in the atmosphere.

Part A: Recharging Raindrops (4.6 points)

The first thought that arises about the charge of raindrops is that when they fall, they retain the same charge that they carried inside the clouds, that is, on average, equal to zero. Experience shows that this assumption is incorrect, and the average charge of drops falling on the Earth's surface is positive. This part of the problem is devoted to the explanation and quantitative analysis of this phenomenon.

You probably know Ohm's law in differential form:

$$\vec{j} = \sigma \vec{E},$$

где $\vec{j}$ и $\vec{E}$ — плотность тока и напряжённость электростатического поля в среде, а σ — величина, обратная удельному сопротивлению и называемая проводимостью. В атмосферном воздухе величина плотности тока обусловлена движением положительно и отрицательно заряженных ионов, движение которых происходит независимо друг от друга, в силу чего можно написать:

$$\vec{j} = \vec{j}_+ + \vec{j}_-,$$

где $\vec{j}_+$ и $\vec{j}_-$ — плотности тока, обусловленные движением положительно и отрицательно заряженных ионов, поэтому закон Ома в дифференциальной форме для атмосферы записывается следующим образом:

$$\vec{j} = \sigma_+ \vec{E} + \sigma_- \vec{E}.$$

Здесь σ_+ характеризует плотность тока, обусловленную движением положительно заряженных ионов, а σ_- — плотность тока, обусловленную движением отрицательно заряженных ионов. При этом σ_+ и σ_- зависят от концентрации ионов. Следует, что положительно заряженные ионы движутся вдоль направления электрического поля, а отрицательно заряженные ионы — противоположно ему. Разница между положительно и отрицательно заряженными ионами в атмосфере заключается не только в отличии знаков их зарядов, но и в их концентрациях. В Земной атмосфере преобладают положительные ионы, поэтому для неё всегда будем считать выполненным соотношение:

$$\sigma_+ > \sigma_-.$$

Перейдём непосредственно к постановке задачи. Водяная капля, которую можно считать проводящим шаром радиусом R , расположена в направленном вертикально вниз электростатическом поле напряжённостью $\vec{E}_0$. Капля движется в атмосфере с положительными и отрицательными ионами, проводимости которых равны σ_+ и σ_- соответственно. Глобальной целью данной части задачи является нахождение стационарного заряда капли Q_0 . При достаточно быстром падении капли по отношению к окружающему воздуху изменение заряда шара обусловлено только движением ионов к поверхности капли.

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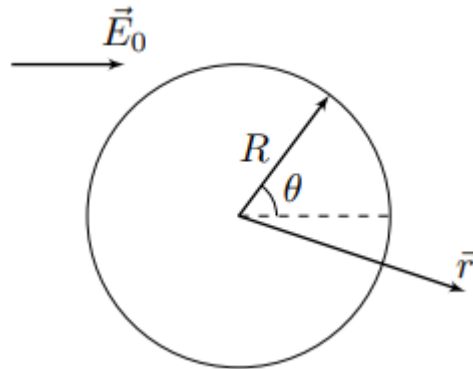


Fig. 1.

When solving, use the following model: When falling, a drop always retains the shape of a sphere with radius R ; The magnitude and direction of the electrostatic field $\vec{E}_0$ remain constant throughout the entire fall of the drop; The charge of a drop changes only when it comes into contact with positively and negatively charged ions, and the charges already on the surface of the drop never leave it.

When solving, use the following model:

- A1** Let a conducting ball of radius R , carrying a charge Q , be placed in a uniform electrostatic field of intensity $\vec{E}_0$. Determine the total strength $\vec{E}$ of the electric field at a point with radius vector $\vec{r}$ relative to the center of the ball, located outside the ball. Express your answer in terms of Q , $\vec{E}_0$, R , ε_0 and $\vec{r}$. **0.80**

Let $\vec{n}$ be the normal vector to the surface of the ball, directed outward.

- A2** Let θ be the angle between the direction of the electrostatic field vector $\vec{E}_0$ and the radius vector $\vec{r}$ of some point on the surface of the ball relative to its center. Determine the projection of the electric field strength $E_n(\theta)$ onto the normal direction. Express your answer in terms of Q , E_0 , R , ε_0 and θ . **0.40**

- A3** At what values of the charge Q of the ball can the value E_n vanish? Express your answer in terms of E_0 , ε_0 and R . Further, at all points, consider that the charge of the ball Q falls within the range you found. **0.20**

Let's move on to analyzing the change in the charge of the ball. Consider some element of its surface dS with a normal vector $\vec{n}$. If at the considered point on the surface of the ball the electrostatic field strength is equal to $\vec{E}$, then per unit time a charge dq/dt is received on this surface area, equal to:

$$\frac{dq}{dt} = \begin{cases} -\sigma_- E_n dS & \text{при } E_n > 0 \\ -\sigma_+ E_n dS & \text{при } E_n < 0 \end{cases}$$

Совпадение знаков в обоих случаях обусловлено тем, что при $E_n > 0$ на поверхность шара попадают отрицательные ионы, а при $E_n < 0$ - positive.

A4 Let the charge of the ball be equal to Q . At what value of the angle θ_0 does the tension component $E_n(\theta_0)$ become zero? Express your answer in terms of Q , E_0 , ε_0 and R . Also determine at what values of the angle θ negative ions hit the surface of the ball, and at what values – positive ones. Express your answers using θ_0 . **0.50**

A5 Determine the total time derivative of the charge of the ball dQ/dt . Express your answer in terms of Q , E_0 , ε_0 , R , σ_+ , σ_- and, if necessary, θ_0 . **1.00**

A6 Determine the stationary charge of the ball Q_0 at which it remains constant in time. Express your answer in terms of E_0 , ε_0 , R , σ_+ and σ_- . **0.80**

Let $\Delta Q = Q - Q_0$ be the deviation of the ball charge from the stationary one. At the moment of time $t = 0$ the deviation of the ball charge from the stationary one was $\Delta Q_0 = Q(0) - Q_0$, and $|\Delta Q_0| \ll Q_0$.

A7 For small values of ΔQ the dependence $\dot{Q}(\Delta Q)$ can be represented as: **0.60**

$$\dot{Q} \approx A \Delta Q.$$

Выразите A через σ_+ , σ_- и ε_0 . Является ли найденное значение заряда Q_0 stable? Justify your answer.

A8 Obtain the dependence of the ball charge deviation ΔQ on time t . Express your answer in terms of ΔQ_0 , ε_0 , σ_+ , σ_- and t . Show also that the type of time dependence is determined only by the product $\sigma_+ \sigma_-$. **0.30**

Next we will study a complex of phenomena called thunderstorms. It covers a whole range of issues related to both thunderstorm discharges, that is, lightning, and glow discharges associated with the corona of pointed objects.

Part B. Occurrence of corona discharge at the tip (2.4 points)

Let us consider the issue of an anomalous increase in the strength of the electrostatic field near pointed objects. Let the conductor be an elongated ellipsoid of revolution, given by the equation:

$$\frac{z^2}{a^2} + \frac{x^2 + y^2}{b^2} = 1.$$

Здесь a и b обозначают большую и малую полуоси эллипсоида соответственно. Ось z эллипсоида ориентирована параллельно внешнему однородному электростатическому полю напряжённостью $\vec{E}_0$, направленному вертикально вниз. Можно показать, что напряжённость $\vec{E}$ электрического поля внутри изолированного равномерно заряженного по объёму эллипсоида вращения зависит линейно от координат x, y, z :

$$\vec{E} = \frac{\rho}{\varepsilon_0} (Az\vec{e}_z + B(x\vec{e}_x + y\vec{e}_y)).$$

Здесь A и B

$$A = \left(\frac{b}{c}\right)^2 \left(\frac{a}{c} \cdot \ln \frac{a+c}{b} - 1\right) \quad B = \frac{a}{2c} \left(\frac{a}{c} - \left(\frac{b}{c}\right)^2 \ln \frac{a+c}{b}\right),$$

где $c = \sqrt{a^2 - b^2}$ [M14] $b \ll a$, то указанные выражения можно приблизить следующими:

$$A \approx \left(\frac{b}{a}\right)^2 \left(\ln \frac{2a}{b} - 1\right) \quad B \approx \frac{1}{2} \left(1 - \left(\frac{b}{a}\right)^2 \ln \frac{2a}{b}\right)$$

Далее во всех пунктах, требующих подстановки A , используйте приближённое выражение.

Задача о помещении проводящего эллипсоида в однородное электрическое поле, направленное параллельно его оси z , может быть решена аналогично задаче для проводящего шара: Рассмотрим два эллипсоида с совпадающей осью z , несущих заряды с постоянными плотностями ρ и $-\rho$. Центр положительного эллипсоида имеет радиус-вектор $\vec{l}$ относительно центра отрицательно заряженного эллипсоида.

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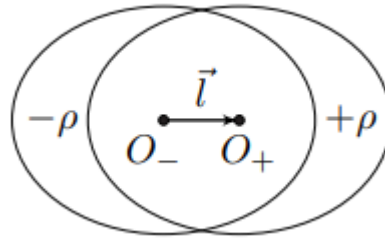


Fig. 2.

- B1** Determine the vector $\vec{E}$ of the electrostatic field strength in the area of intersection of isolated ellipsoids. Express your answer in terms of ρ , $\vec{l}$, ε_0 and A . **0.50**

If the distance l between the centers of the ellipsoids tends to zero, then their combination can be considered as one ellipsoid with a constant polarization vector $\vec{P}$.

- B2** Let us consider an isolated ellipsoid uniformly polarized along the z axis. Let the electrostatic field strength inside the ellipsoid be equal to $\vec{E}$. Determine the polarization vector of the ellipsoid $\vec{P}$. Express your answer in terms of $\vec{E}$, ε_0 and A . **0.30**

Let us consider an uncharged conducting ellipsoid placed in a uniform electrostatic field of intensity $\vec{E}_0$, directed along the axis z of the ellipsoid.

- B3** Determine the polarization vector $\vec{P}$ of the ellipsoid. Express your answer in terms of $\vec{E}_0$, ε_0 and A . Also determine the maximum value of the surface charge density σ_{max} on the surface of the ellipsoid. Express your answer using P . **0.30**

- B4** Express the total component of the electrostatic field strength E_n on the surface of the conductor in terms of the surface charge density σ and ε_0 . Determine the maximum value of the electrostatic field strength E_{max} on the surface of the ellipsoid. Express your answer in terms of E_0 and A . **0.40**

Let us consider half of an ellipsoid uniformly polarized along the z axis as shown in the figure below. With its equatorial cross-section it is in contact with an infinite conducting plane.

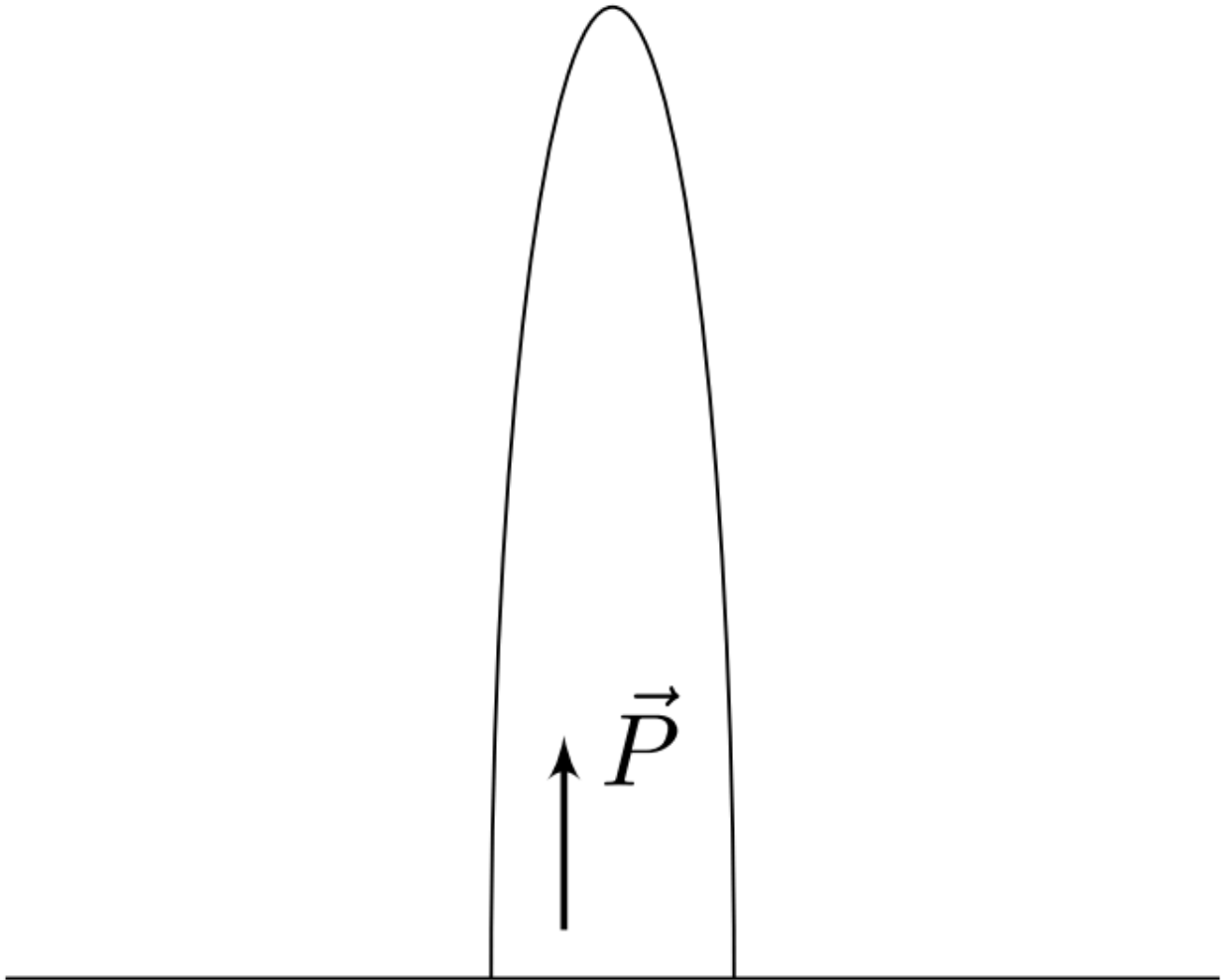


Fig. 3.

- B5** The answer sheets show a drawing in which a point dipole is located above an infinite conducting plane, the dipole moment of which is directed perpendicular to the plane. In the answer sheets, provide the electrostatic image of a dipole in a conducting plane. Using the result obtained, provide in the answer sheets an electrostatic image of half of a uniformly polarized ellipsoid of revolution in contact with a conducting plane with an equatorial section. **0.40**

The above reasoning can be used when studying a real structure: Let us consider a conductor in the form of half an elongated ellipsoid of revolution with semi-axes $a = 1\text{km}$ and $b = 1\text{mm}$, resting with its equatorial cross-section on the horizontal surface of the Earth, which can also be considered conductive. In the atmosphere, at a great distance from the ellipsoid, there is a vertically downward electrostatic field of strength $E_0 = 100\text{B/M}$.

- B6** Show that the expression for the maximum electrostatic field strength E_{max} is the same as the expression found in B4 and find its numerical value. Is the magnitude of the electrostatic field strength E_0 sufficient for the breakdown of air at any point in space if it occurs at a strength equal to $E_{np} = 30 \text{ kV/cm}$? **0.50**

Part C. Extraction of electricity from the atmosphere (1.6 points)

Using a design similar to that considered in part B of the problem, it is possible to achieve the extraction of electric charge from the atmosphere. In this part of the problem, the conductivity of atmospheric air under normal conditions is equal to σ_0 , and the effects described in part A of the problem can be neglected. Consider a conducting ball of radius R_0 connected to the Earth by a long thin straight wire of length $h \gg R_0$. Consider the radius of the wire small compared to R_0 . The electrostatic field in the Earth's atmosphere can be assumed to be constant, directed vertically downward and equal to E_0 .

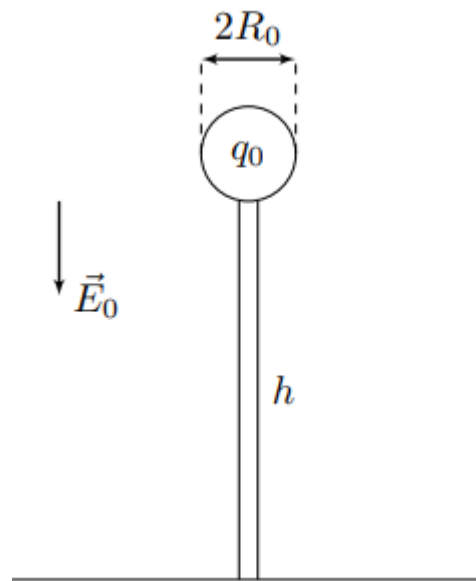


Fig. 4.

- C1** Taking the potential of the ball equal to the potential on the surface of the Earth, i.e. zero, determine the magnitude of the charge q_0 of the ball. Express your answer in terms of ϵ_0 , E_0 , R_0 and h . The influence of the electrostatic field of the ball on the electrostatic field of the Earth can be neglected. The influence of the electrostatic field of charges located on the wire can be neglected throughout the entire space. **0.30**
- C2** Determine the magnitude of the current I flowing from the atmosphere into the ball if the conductivity of air in the entire atmosphere can be taken equal to σ_0 . Express your answer in terms of E_0 , R_0 , h , ϵ_0 and σ_0 . **0.40**

From the results obtained in the previous paragraphs of parts B and C it is clear that the magnitude of the electrostatic field strength near the surface of the conductor can reach quite large values, which leads to ionization of the layers of air closest to it and a strong increase in their conductivity. To take this effect into account, we will assume that the conducting ball is located inside a spherical shell with internal and external radii R_0 and R , respectively, in which ionized air with conductivity σ is located. At distances $r > R$ from the center of the ball, the conductivity remains equal to σ_0 . Consider the distribution of charges in the medium to be stationary, so that at any point in space the charge density remains constant. In this system, the charge is

located only on spherical surfaces with radii R_0 and R , and on the surface with radius R_0 the total charge is q_0 , and on the surface with radius $R - q$ (see figure).

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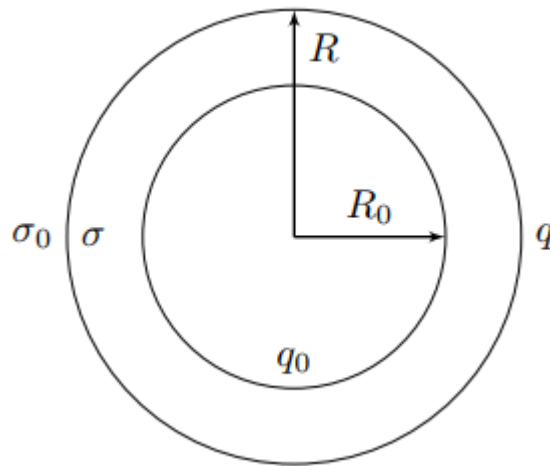


Fig. 5.

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|-----------|---|-------------|
| C3 | Write down the expression for the condition that the ball's potential is equal to zero. The equations may include E_0 , h , q_0 , q , R_0 and R . | 0.30 |
| C4 | From the condition of equality of the strength of the current I crossing the spherical surface inside and outside the ionized layer, obtain an equation relating the charges q_0 and q . The conductivities σ_0 and σ can also enter into the equation. | 0.30 |
| C5 | Determine the magnitude of the current I flowing from the atmosphere into the ball in this case. Express your answer in terms of E_0 , R_0 , R , h , ε_0 , σ_0 and σ . | 0.20 |
| C6 | Show that at $R \approx R_0$ the approximate expression for the current I becomes an expression corresponding to the absence of an ionized layer. | 0.10 |

Part D. Lightning propagation channel (2.4 points)

We investigate the formation and development of a lightning discharge. As an example, consider the mechanism of lightning formation from the tip considered in the B part of the problem. A conductor can create a strong electrostatic field near itself, causing ionization of the air. Ionized air behaves as a conductor for a relatively long time, since the process of recombination of ions in it occurs slowly. As a result of this, the conducting region of space will grow until the electrostatic field is sufficient for ionization. If the conducting region (spark channel) formed in this way connects the clouds and the surface of the Earth, then a

very large electric current can flow through it. This part of the problem is devoted to estimating the rate of formation of the spark channel. It was experimentally found that it is approximately equal to the speed of movement of free electrons at the boundary of the conducting region, where the field strength is equal to the critical value E_{np} at which ionization occurs.

The proposed model of the phenomenon is as follows: In the atmospheric region there are electrons with a charge $-e$, the mass of which is equal to m . Also in the atmosphere there are neutral air molecules, the mass M of which is many times greater than the mass of electrons, i.e. $M \gg m$. Under the influence of an electrostatic field, electrons become accelerated and collide with air molecules, and the average distance traveled by electrons between two successive collisions with air molecules is λ . The quantity λ is called the free path. In this case, air molecules can always be considered motionless. The average kinetic energy of thermal motion of electrons is equal to $\overline{W}$ and is so great that between two successive collisions there is practically no time to change. Consider that the value $\overline{W}$ is related to the average value of the speed of thermal motion of electrons $\overline{v}_T$ by the relation:

$$\overline{W} = \frac{m\overline{v}_T^2}{2}.$$

Вся данная конструкция расположена в однородном электрическом поле $\vec{E}$, направленном противоположно скорости распространения искрового канала.

Средний вектор тепловой скорости теплового движения электронов равен нулю:

$$\langle \vec{v}_T \rangle = 0.$$

Тогда среднее перемещение электронов в направлении электростатического поля за время $\tau = \lambda/\overline{v}_T$, соответствующее характерному времени движения электрона между двумя последовательными столкновениями с молекулами воздуха, составляет:

$$\langle \vec{S} \rangle = \langle \vec{v}_T \rangle \tau + \vec{a} \tau^2 / 2 = \vec{a} \tau^2 / 2 \Rightarrow \vec{u} = \frac{\langle \vec{S} \rangle}{\tau} = \frac{\vec{a} \tau}{2}.$$

Вектор $\vec{u}$ is called the average electron drift velocity in electrostatic field. This value is equal to the speed of propagation of the spark channel.

D1 Determine the value of the drift velocity u of electron motion. Express your answer in terms of e , E , λ , m and $\overline{v}_T$. **0.20**

When the average value of the kinetic energy of thermal motion of electrons reaches a stationary value $\overline{W}$, the average power of loss of kinetic energy of electrons when colliding with air molecules is compensated by the work of the electric field to move electrons.

D2 From the energy balance, determine the average value $\overline{\Delta W}$ of the loss of kinetic energy of an electron upon collision with air molecules. Express your answer in terms of e , E , λ , m and $\overline{W}$. **0.40**

The quantity $\overline{\Delta W}$ can also be found directly from collision analysis.

To determine the value of $\overline{\Delta W}$ we accept the following model: All collisions occurring between electrons and air molecules are elastic; Initially, an air molecule with mass M is at rest, and the kinetic energy of an

electron with mass m incident on it is always equal to $\overline{W}$; In the reference frame of the center of mass of the considered system of two particles, all directions of the speed of an air molecule immediately after a collision with an electron are equally probable. Let φ be the angle between the velocity vector of the incident electron and the velocity of the air molecule in the reference frame of the center of mass immediately after the collision. Then, since all directions of the speed of an air molecule in the reference frame of the center of mass are equally probable, the average value of the electron kinetic energy loss is determined by the expression:

$$\overline{\Delta W} = \frac{1}{4\pi} \int_{\Omega} E_k(\varphi) d\Omega,$$

где $E_k(\varphi)$ – кинетическая энергия в лабораторной системе отсчёта молекулы воздуха, сразу после столкновения движущейся в направлении угла φ в системе отсчёта центра масс, а $d\Omega = 2\pi \sin \varphi d\varphi$ – телесный угол, под которым виден сегмент поверхности сферы между углами φ и $\varphi + d\varphi$.

Пусть φ – угол между вектором скорости налетающего электрона и скоростью молекулы воздуха в системе отсчёта центра масс сразу после столкновения. Тогда, поскольку все направления скорости молекулы воздуха в системе отсчёта центра масс равновероятны, среднее значение потерь кинетической энергии электрона определяется выражением:

$$\overline{\Delta W} = \frac{1}{4\pi} \int_{\Omega} E_k(\varphi) d\Omega,$$

где $E_k(\varphi)$ – кинетическая энергия в лабораторной системе отсчёта молекулы воздуха, сразу после столкновения движущейся в направлении угла φ в системе отсчёта центра масс, а $d\Omega = 2\pi \sin \varphi d\varphi$ – телесный угол, под которым виден сегмент поверхности сферы между углами φ и $\varphi + d\varphi$.

D3 Obtain an exact expression for the quantity $\overline{\Delta W}/\overline{W}$. Express your answer in terms of m and M . Simplify your answer to include $m \ll M$. If you were unable to solve this point, in the future consider that $\overline{\Delta W}/\overline{W} = m/M$. **1.00**

Consider the following data known: Electron charge $-e = -1.6 \cdot 10^{-19}$ Кл; Electron mass $m = 9.1 \cdot 10^{-31}$ кг; Mass of an air molecule $M \approx 6 \cdot 10^{-26}$ кг; The critical electrostatic field strength at which breakdown occurs is $E_{\text{пр}} = 30$ кВ/см; Average free path $\lambda \approx 1$ мкм.

Consider the following information known:

D4 Determine the stationary value of the kinetic energy of thermal motion of electrons $\overline{W}$ and the rate of formation of the spark channel u . Express your answers in terms of m , M , e , E and λ . Calculate the resulting values. **0.80**

Source: <https://pho.rs/p/3703>